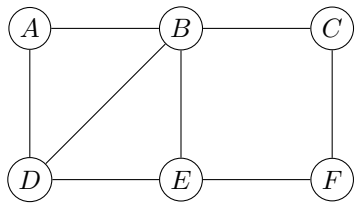
Phase 3 (part 1): Network Basics

1.] Consider the undirected graph below.



a.) Write the vertex set V and edge set E. Note: E is also the *edge list*.

$V = \{A, B, C, D, E, F\}$

$E = \{(A,B), (A,D), (B,C), (B,D), (B,E), (C,F), (D,E), (E,F)\}$

b.) How many vertices and edges does the graph have?

6 vertices $|V| = 6$

8 edges $|E| = 8$

c.) Compute the degree of nodes A and B.

$\deg(A) = 2$ $\deg(B) = 4$

d.) Write the *adjacency list* of the graph.

$N(A) = \{B, D\}$ $N(C) = \{B, F\}$ $N(E) = \{B, D, F\}$

$N(B) = \{A, C, D, E\}$ $N(D) = \{A, B, E\}$ $N(F) = \{C, E\}$

e.) Fill in the *adjacency matrix* A where $A_{ij} = 1$ if there is an edge between the vertices and 0 otherwise. What do you notice about this matrix? How can we compute $\deg(B)$ from this matrix?

	A	B	C	D	E	F
A	0	1	0	1	0	0
B	1	0	1	1	1	0
C	0	1	0	0	0	1
D	1	1	0	0	1	0
E	0	1	0	1	0	1
F	0	0	1	0	1	0

$\deg(B) =$ row sum or column sum $= \boxed{4}$

f.) Create this graph in Python using an edge list, adjacency list, and the adjacency matrix.

bemerich_p3ws1_1f.py

2.] Suppose the same graph is now weighted, with edge weights shown below:

Edge	Weight
$A-B$	3
$B-C$	2
$A-D$	4
$B-E$	5
$C-F$	1
$D-E$	2
$E-F$	3
$B-D$	2

$\longrightarrow$ Adjacency matrix

$$\begin{array}{c|cccccc} & A & B & C & D & E & F \\ \hline A & 0 & 3 & 0 & 4 & 0 & 0 \\ B & 3 & 0 & 2 & 2 & 5 & 0 \\ C & 0 & 2 & 0 & 0 & 0 & 1 \\ D & 4 & 2 & 0 & 0 & 2 & 0 \\ E & 0 & 5 & 0 & 2 & 0 & 3 \\ F & 0 & 0 & 1 & 0 & 3 & 0 \end{array}$$

a.) Create this graph in Python using an edge list with the corresponding weights.

b.) Compute the *strength* of vertex B.

$S(A) = (\text{row } 1 \text{ sum})$ or $(\text{col } 1 \text{ sum}) = \boxed{7}$

$S(B) = 3+2+5+2 = \boxed{12}$

$S(C) = 3 \quad S(E) = 10$

$S(D) = 8 \quad S(F) = 4$

c.) Using weights, which path from A to F has smaller total weight? Compare these two paths by total weight:

$$A \to B \to E \to F \qquad \text{and} \qquad A \to D \to E \to F.$$

Path: $\underbrace{A \to B}_{3} \underbrace{\to E}_{5} \underbrace{\to F}_{3} \longrightarrow$ weight $= 11$

Path: $\underbrace{A \to D}_{4} \underbrace{\to E}_{2} \underbrace{\to F}_{3} \longrightarrow$ weight $= 9$

Path $A \to D \to E \to F$ has smaller total weight

3.] Use Python to solve parts b and c above. Then, find the *shortest path* from node A to node F.

bemerick_p3ws1_3.py